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Separating effects of clonal integration on plant growth during submergence and de-submergence

Guan-Wen Wei[a], Qi Shu[a], Fang-Li Luo[a,*], Yu-Han Chen[a], Bi-Cheng Dong[a], Li-Chun Mo[a], Wen-Jun Huang[b], Fei-Hai Yu[c,d]

[a] School of Nature Conservation, Beijing Forestry University, Beijing, 100083, China
[b] Sichuan Academy of Forestry, Chengdu, 610081, China
[c] Institute of Wetland Ecology & Clone Ecology, Taizhou University, Taizhou, 318000, Zhejiang, China
[d] Zhejiang Provincial Key Laboratory of Plant Evolutionary Ecology and Conservation, Taizhou University, Taizhou, 318000, Zhejiang, China

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ABSTRACT

Responses to both submergence and de-submergence are important for evaluating flood tolerance of plants. Clonal integration (resource translocation between connected ramets within clones) has been shown to increase flood tolerance of amphibious clonal plants. However, no study has truly separated the effects of clonal integration during de-submergence from those during submergence. We grew 40 clonal fragments of the amphibious plant species *Alternanthera philoxeroides*, each consisting of two ramet systems, under half-submerged conditions (one ramet system submerged and the other un-submerged) for 30 d and then de-submerged for 20 d. To evaluate the effects of clonal integration during submergence, stolon connection between the two ramet systems was either severed or not on day 1, and 10 replicates of the fragments of each treatment were harvested on day 30. To evaluate the effects of clonal integration after de-submergence, the remaining 20 clonal fragments with connected stolons during submergence were de-submerged; half of them were subjected to stolon severing on day 31 and the remaining 10 fragments were not. All the 20 fragments for de-submergence were harvested on day 50. After 30 d of submergence, stolon connection between the submerged and the un-submerged ramets significantly increased growth, biomass allocation to roots and photosynthetic capacities of the submerged ramets, and also increased growth and photosynthetic capacities of the un-submerged ramets. Consequently, clonal integration significantly increased growth of the apical ramets, the basal ramets and the whole fragments. However, after 20 d of de-submergence, stolon connection did not significantly affect growth or photosynthetic capacities of either de-submerged or un-submerged ramets. The results suggest that clonal integration plays different roles during submergence and de-submergence. It increases performance of submerged ramets during submergence by translocation of photosynthates from un-submerged ramets, but plays little roles during de-submergence.

1. Introduction

Submergence and de-submergence are common stresses encountered by riparian plants due to water level fluctuations (McGowan et al., 2011; Raulings et al., 2010). Both of them can strongly influence growth and physiological processes of plants (Striker and Colmer, 2017; Voesenek and Bailey-Serres, 2015). Submergence, especially under turbid floodwater, greatly decreases gas diffusion and light intensity underwater, leading to carbohydrate starvation and fatal injury of plants (Voesenek et al., 2006; Wang et al., 2016). After de-submergence, plant injuries developed underwater can be greatly intensified (Striker, 2012). Sudden increase in O_2 and light intensity may lead to over-accumulation of reactive oxygen species and toxic oxidative products in plants (Blokhina et al., 2003; Santosa et al., 2007). Besides, low hydraulic conductivity of roots is not able to provide enough water to meet shoot transpirational demand, causing wilting of shoots in many plant species (Holbrook and Zwieniecki, 2003). Therefore, plant injuries suffered from de-submergence can be as serious as those suffered from submergence (Malik et al., 2002; Striker, 2012).

Submergence-tolerant species often can alter morphology and anatomy of submerged shoots and roots to facilitate internal diffusion of gases such as O_2, CO_2 and ethylene (Colmer and Voesenek, 2009; Luo et al., 2016; Mommer and Visser, 2005). Some of them can also adjust

* Corresponding author.
E-mail address: ecoluofangli@bjfu.edu.cn (F.-L. Luo).

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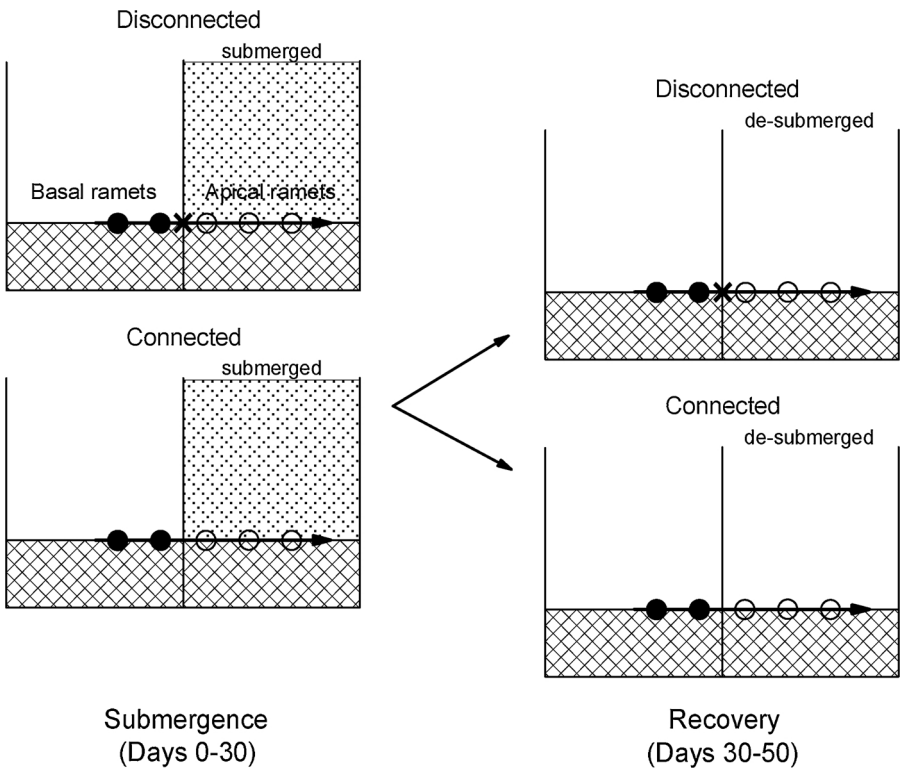

Fig. 1. Schematic representation of the experimental design. Each clonal fragment of *Alternanthera philoxeroides* consists of two basal ramets (closed circles) and three apical ramets (open circles) with a stolon apex (arrow). During days 0–30, the apical ramets were submerged and the basal ramets were un-submerged. On day 31, the apical ramets were de-submerged. The basal and the apical ramets were either connected or disconnected by severing to evaluate roles of clonal integration during submergence and de-submergence.

Table 1
The split-plot ANOVA results for effects of clonal integration (I), section (S) and their interaction on growth and chlorophyll fluorescence of *Alternanthera philoxeroides* during submergence and de-submergence.

	Submergence			De-submergence		
	I	S	I × S	I	S	I × S
Leaf biomass	2.40	**4.64***	0.98	0.01	0.55	1.01
Stem biomass	**8.83****	0.25	1.11	0.31	0.11	0.67
Root biomass	1.67	**7.42***	**4.79***	0.03	0.02	1.03
Total biomass	**4.66***	1.72	0.24	0.15	0.01	0.87
Root to shoot ratio	1.22	**4.12**$^{\#}$	**5.45***	0.11	0.40	0.21
Main stem length	**3.48**$^{\#}$	**23.73*****	0.40	0.39	**14.71****	0.30
Ramet number	**7.66***	0.47	0.01	2.22	**50.12*****	1.69
Fv/Fm	0.08	**28.18*****	0.29	0.21	2.63	0.23
ΔF/Fm'	**16.87****	**32.96*****	2.20	0.23	1.83	0.23
NPQ	1.61	**16.17****	0.27	1.03	0.60	0.09

Values are *F* and symbols show p (***, $p < 0.001$; **, $p < 0.01$; *, $p < 0.05$; $^{\#}$, $0.05 \le p < 0.1$ and no symbols $p \ge 0.1$). Degrees of freedom for effects of I, S and I × S are (1, 18), (1, 18) and (1, 18), respectively. For the analyses, the whole clonal fragments are as main plots and the apical/basal ramets are as subplots.

metabolic pathways by switching from aerobic to anaerobic respiration to alleviate energy crisis (Gibbs and Greenway, 2003). When floodwater is subsided, some of these species have the ability of delaying chlorophyll degradation (Roiloa and Retuerto, 2016), retaining functional PSII complexes even after prolonged submergence (Herrera, 2013), or recovering growth and photosynthetic capacities quickly (Luo et al., 2011).

The ability of plants to tolerate submergence is determined not only by their performance during submergence, but also by their performance after de-submergence (Striker, 2012; Voesenek and Bailey-Serres, 2015). Several studies have shown that performance of plants after de-submergence was not necessarily positively related to that during submergence (Chen et al., 2011; van Eck et al., 2004; Striker et al., 2011). For example, species showing poor performance during submergence may not be submergence-sensitive because they can quickly resume growth by using reserved carbohydrates and/or newly-formed photosynthates after de-submergence (Akman et al., 2012; Striker et al., 2012). However, plant responses after de-submergence have often been overlooked (Malik et al., 2002; Striker, 2012). Therefore, the conclusions on plant tolerance to submergence have been mainly based on their responses during or immediately after submergence. In recent years, increasing number of studies have investigated effects of de-submergence on recovery of plant growth and photosynthetic capacities (van Eck et al., 2004; Gautam et al., 2015; Luo et al., 2011, 2014; Malik et al., 2002; Striker, 2012). However, the 'de-submergence effects' evaluated in these studies in reality tested those on the responses of plants to the integrated effects of both submergence and de-submergence.

Many amphibious species are clonal plants, which enable them to spread their asexual individuals (ramets) from terrestrial to aquatic habitats and *vice versa* (Lin et al., 2017; Liu et al., 2016; Wang et al., 2009). The connecting stolons between ramets within a clone allow the translocation of resources such as carbohydrates, water and nutrients between connected ramets, i.e., clonal integration (Roiloa et al., 2014; Song et al., 2013; Zhou et al., 2017). Many studies have shown that clonal integration increases performance of ramets suffering from various environmental stresses in both terrestrial and aquatic conditions (Liu et al., 2016; Luo et al., 2014; Roiloa et al., 2014; Wang et al., 2017a,b; Yu et al., 2004, 2008). Several studies on roles of clonal integration in resisting flooding stress have shown that clonal integration benefits submerged ramets with the translocation of photosynthates and probably also O_2 from un-submerged ramets during the submergence (Ayi et al., 2016; Luo et al., 2014; Wang et al., 2009; Xiao et al., 2010).

In this study, we used a novel design to truly separate effects of clonal integration during submergence and de-submergence with the well-studied amphibious clonal plant, *Alternanthera philoxeroides*. The main aim is to test whether clonal integration affects growth and photosynthetic capacities of *A. philoxeroides* after de-submergence. We also address whether clonal integration increases performance of *A. philoxeroides* during submergence. Our hypothesis is that clonal

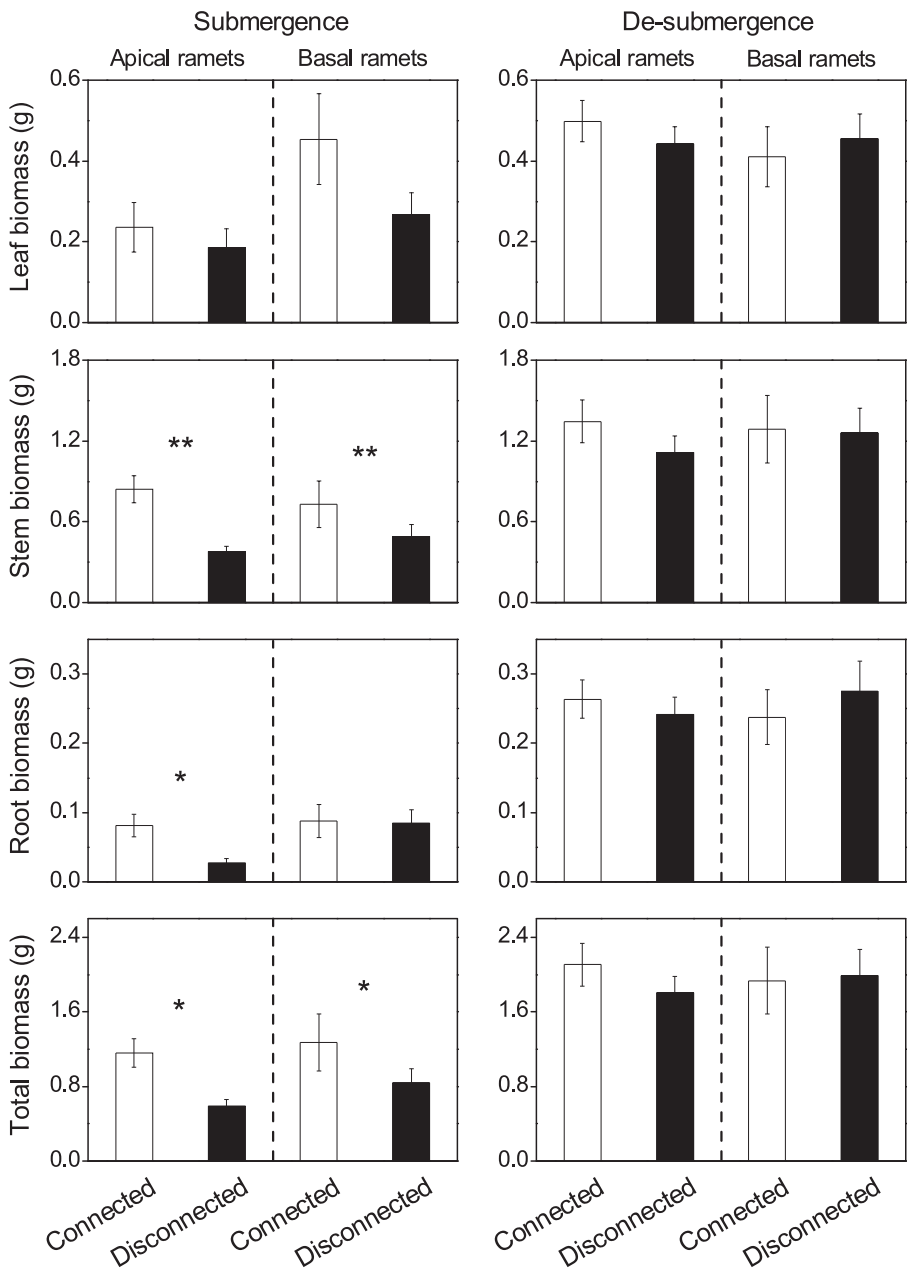

Fig. 2. Leaf biomass, stem biomass, root biomass and total biomass of the apical and the basal ramets of *Alternanthera philoxeroides* after submergence (day 30) and the subsequent recovery (day 50). Symbols represent statistically significant differences between the connected and the disconnected treatments of the apical or the basal ramets (**, $p < 0.01$; *, $p < 0.05$). Data are mean values (± s.e., $n = 10$).

integration can increase performance of clonal plants during both submergence and de-submergence.

2. Material and methods

2.1. Plant species

Alternanthera philoxeroides (Mart.) Griseb. (Amaranthaceae) is an amphibious weed that occurs frequently in riparian regions in China and can establish in aquatic, semi-aquatic and terrestrial environments (Liu and Dong, 2017; Wang et al., 2008, 2009). It originates from South America, and is listed as a highly invasive species in many countries, including China. *Alternanthera philoxeroides* does not produce viable seeds in China and mainly reproduces asexually via its stolon and root fragments. Thus, genetic diversity of *A. philoxeroides* is very low in China. A ramet consists of a stolon node with two leaves, two axillary buds and some roots. This species can form long hollow stems and quickly spreads by clonal growth. It can spread from terrestrial to aquatic areas and *vice versa*, and quickly elongates its shoots to regain contact with the atmosphere when submerged in deep water (Luo et al., 2011).

2.2. Material preparations

Plants of *A. philoxeroides* were collected from five plots at least 10 m apart in each of two wetlands in Taizhou, Zhejiang Province, China. These plants were mixed and propagated vegetatively in a greenhouse at the Ecological Garden of Beijing SFK Technology Co, Ltd. For the experiment, the clonal fragments of *A. philoxeroides*, each consisting of five nodes (ramets), were at the same developmental stage and propagated from pre-cultivated stem cuttings. The two older ramets were referred to 'basal ramets' and the three younger ramets referred to

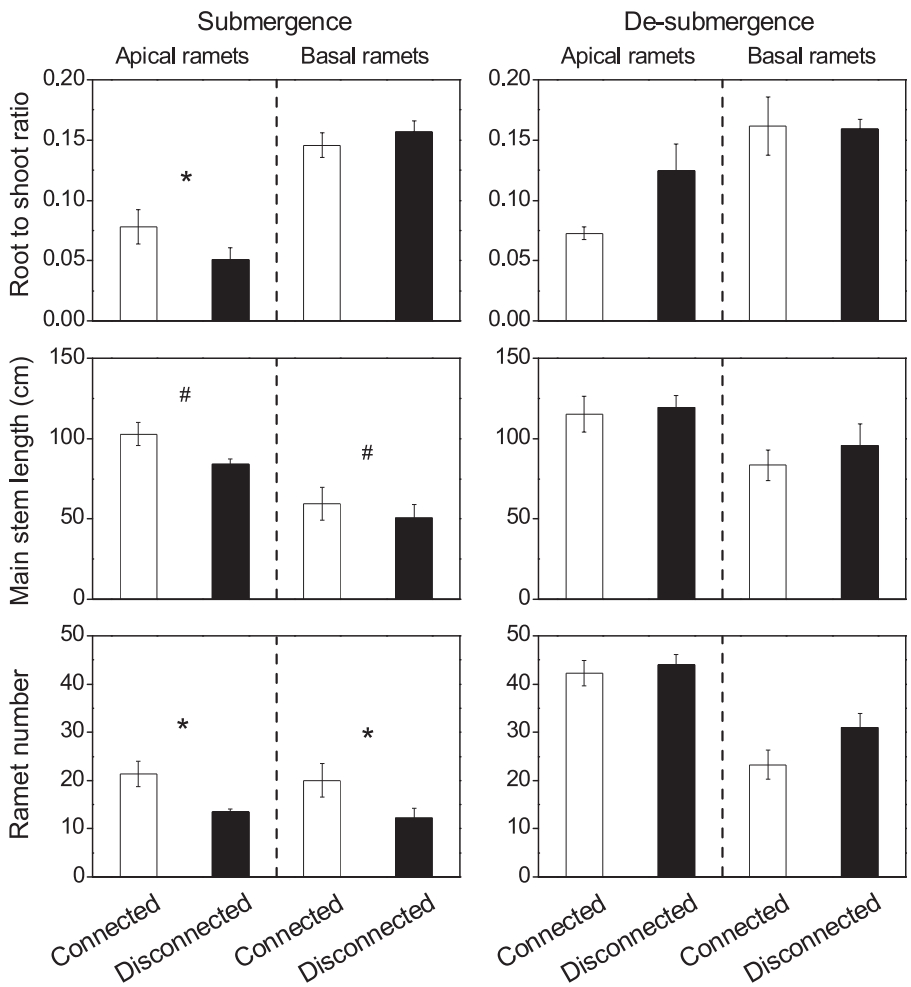

Fig. 3. Root to shoot ratio, main stem length and ramet number of the apical and the basal ramets of *Alternanthera philoxeroides* after submergence (day 30) and the subsequent recovery (day 50). Symbols represent statistically significant differences between the connected and the disconnected treatments of the apical or the basal ramets (*, $p < 0.05$; #, $0.05 \leq p < 0.1$). Data are mean values (± s.e., $n = 10$).

'apical ramets' (Fig. 1). The apical and basal ramets were grown in two adjoining containers (each container: 40 cm × 20 cm × 60 cm, length × width × height). Each container was filled with a 12 L mixture of riverbank soil and sand (2:1, v:v), containing 0.48 ± 0.05 mg total nitrogen g^{-1} and 0.95 ± 0.06 mg total phosphorous g^{-1}. The adjacent containers were divided by a board and a hole (3 cm in diameter) was made in the board 14 cm from the base, allowing the stolon connecting the apical and basal ramets to pass through. For the submergence treatment, tap water was added into the containers. To avoid water flow through the hole between the two adjoining containers, the stolon was enclosed in a rubber stopper that was placed in the hole.

2.3. Experimental design

The experiment was conducted outdoors of the Ecological Garden. On July 22, 2014, 50 plants (clonal fragments) were selected and planted in 50 pairs of adjoining containers. After two weeks of recovery, ten of them were randomly selected and oven dried to measure the initial size of the clonal fragments (0.23 ± 0.01 g, mean ± s.e.). The remaining 40 plants were half-submerged during the first 30 days, i.e., the apical ramets were submerged under 45-cm deep water and covered with a neutral shading cloth which did not change red to far-red ratio, whereas the basal ramets were kept un-submerged, i.e., grown in the soil in water-saturated condition. During the submergence, the shading cloths were fixed 2 cm below the water surface in order to keep the plants completely underwater to mimic the conditions of deep or turbid floodwater. To evaluate the roles of clonal integration during submergence, these plants were randomly subjected to two treatments: ten plants for the disconnected treatment and 30 plants for the connected treatment by either severing the stolon connection between apical and basal ramets or not for 30 days. On day 30, the flooding water was subsided and the shading cloths were removed. The ten plants in the disconnected treatment were harvested and ten plants in the connected treatment were randomly selected and also harvested. To evaluate the roles of clonal integration after de-submergence, the remaining 20 plants with connected stolons were randomly assigned to two treatments: ten plants for the disconnected treatment and the other ten plants for the connected treatment by either severing the connection between apical and basal ramets or not for 20 days of recovery.

Tap water was added to the containers to compensate for water loss due to evaporation. The air temperature and relative humidity ranged between 13 and 36 °C and between 61 and 66%, respectively, during the experiment. The photosynthetic photon flux density of natural light in the garden measured at plant level at noon was 800–1800 μmol photons m^{-2} s^{-1} (Li-250 A photometer, Li-Cor Biosciences, Lincoln, NE, USA).

2.4. Growth measurements

On harvest days (days 30 and 50), we measured main stem length and ramet number of both apical and basal ramets of each plant (clonal

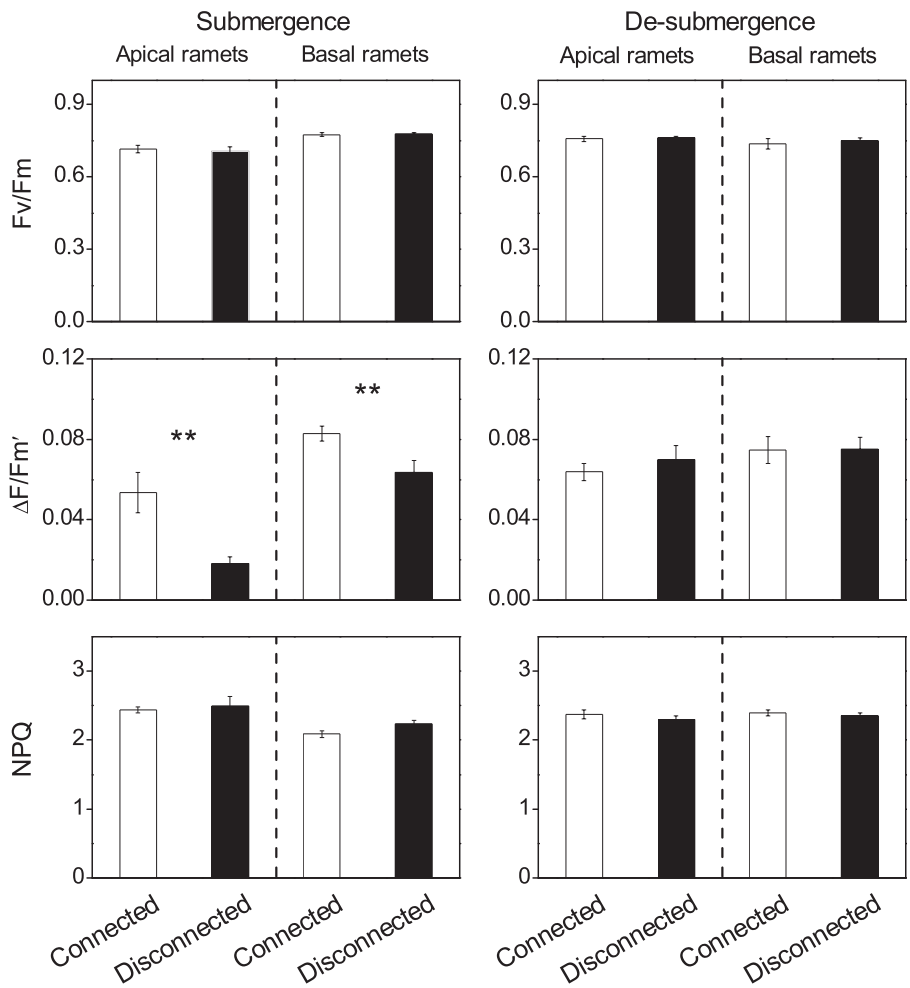

Fig. 4. Maximum quantum yield (Fv/Fm), effective quantum yield (ΔF/Fm′) and non-photochemical energy quenching (NPQ) of the apical and the basal ramets of *Alternanthera philoxeroides* after submergence (day 30) and the subsequent recovery (day 50). Symbols represent statistically significant differences between the connected and the disconnected treatments of the apical or the basal ramets (**, $p < 0.01$). Data are mean values (± s.e., $n = 10$).

fragment). Then leaves, stems and roots of both apical and basal ramets were harvested, washed thoroughly with tap water, dried in an oven at 80 °C at least for 72 h, and weighed.

2.5. Chlorophyll fluorescence measurements

Chlorophyll fluorescence parameters can give insights into photoinhibition, photochemical efficiency and photoprotective energy dissipation of photosystem II (Maxwell and Johnson, 2000). On each harvest day, chlorophyll fluorescence was measured on the first fully expanded leaves of both apical and basal ramets by using a portable modulated fluorimeter (PAM-2500, Heinz Walz, Germany) during 09:00–12:00. The maximal (Fm) and minimal fluorescence intensity (Fo) of dark-adapted leaves were measured after 30 min of dark adaptation with leaf clips. The maximum quantum efficiency of PSII (Fv/Fm) was calculated as: Fv/Fm = (Fm - Fo)/Fm, which is used as a sensitive indicator of photoinhibition (Maxwell and Johnson, 2000). Then, the leaves were exposed to white actinic light at an intensity of 800 μmol m^{-2} s^{-1} for 4 min. The steady-state value of fluorescence (Fs) and the maximal light-adapted fluorescence level (Fm′) were measured by the same procedure. The effective quantum yield of PSII in the light (ΔF/Fm′, an indicator of overall photosynthetic capacity) was calculated as: ΔF/Fm′ = (Fm′ - Fs)/Fm′. Non-photochemical energy quenching in PSII (NPQ) is linearly related to heat dissipation, which is calculated as: NPQ = (Fm - Fm′)/Fm′.

2.6. Statistical analyses

Before analyses, data were checked for normality and homogeneity of variance. Data for biomass of the apical stems and biomass of the whole clonal fragments on day 30 were transformed to the square roots. A split-plot ANOVA, with the whole clonal fragments as main plots and the apical / basal ramets as subplots, was used to test effects of clonal integration (connected vs. disconnected, a fixed factor), section (apical vs. basal, a split-plot factor) and their interaction on growth and chlorophyll fluorescence of *A. philoxeroides* during submergence and de-submergence. All analyses were done using SPSS 16.0 (SPSS, Chicago, IL, USA). Effects were considered to be significant if $p < 0.05$.

3. Results

3.1. Effects of clonal integration during submergence

After 30 d of submergence, clonal integration significantly affected stem biomass, total biomass, ramet number and ΔF/Fm′, and marginally affected main stem length of *A. philoxeroides* (Table 1). Stem biomass, total biomass and ramet number of the apical ramets, the basal ramets and the whole fragments were significantly higher in the connected than in the disconnected treatment (Figs. 2, 3 and 5). ΔF/Fm′ of the apical and the basal ramets was significantly higher, and main stem length of the apical and the basal ramets was marginally higher in the connected than in the disconnected treatment (Figs. 3 and 4). The

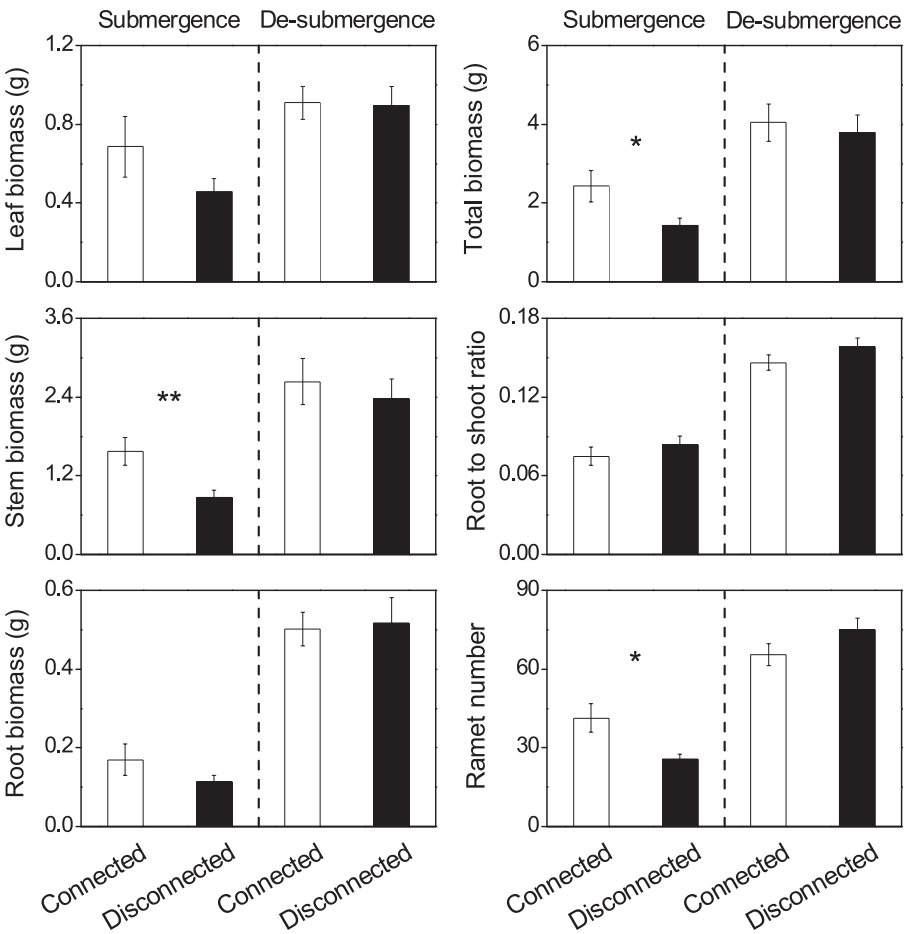

Fig. 5. Leaf biomass, stem biomass, root biomass, total biomass, root to shoot ratio and ramet number of the whole clonal fragments of *Alternanthera philoxeroides* after submergence (day 30) and the subsequent recovery (day 50). Symbols represent statistically significant differences between the connected and the disconnected treatments after submergence or the subsequent recovery (**, $p < 0.01$; *, $p < 0.05$). Data are mean values (± s.e., $n = 10$).

interactive effects of clonal integration and section significantly affected root biomass and root to shoot ratio of plants (Table 1). These two variables of the apical ramets were higher in the connected than in the disconnected treatment, which was not observed in the basal ramets (Figs. 2 and 3).

3.2. Effects of clonal integration during de-submergence

After 20 d of de-submergence, no significant effects of clonal integration on growth or photosynthetic capacities were found between the connected and the disconnected treatments at both ramet and fragment levels (Table 1; Figs. 2–5).

4. Discussion

Clonal integration significantly increased growth and ΔF/Fm′ of submerged apical ramets of *A. philoxeroides*. The results agree with the first part of our hypothesis and suggest that clonal integration can improve the tolerance of ramets to submergence. Similarly, many other studies have also shown the benefits of clonal integration to ramets suffering from various environmental stresses (Roiloa et al., 2014; Song et al., 2013; Wang et al., 2017a,b). We also found positive effects of clonal integration on stem growth and ramet number of the basal ramets. As ΔF/Fm′ of both apical and basal ramets was significantly higher in the connected than in the disconnected treatment, benefits of clonal integration found in the basal ramets in this study arose probably because only surplus resources of the basal ramets were transported to the apical ramets (Liu et al., 2016; Roiloa et al., 2014; Song et al., 2013).

With stolon connection, the submerged apical ramets invested more biomass in roots, while the un-submerged basal ramets invested more biomass in shoots compared with the disconnected treatment. Benefits of clonal integration to the basal ramets with higher allocation to shoots could increase the surface area of photosynthetic organs. Similarly, a flexible biomass allocation was also found in three varieties of this species (Jia et al., 2010), which facilitates plants to maximize the surface area for uptake of the most limiting resources upon stressed conditions (Fan et al., 2015; Liu and Dong, 2017; Poorter and Nagel, 2000). Together with higher ΔF/Fm′, the basal ramets with stolon connection could produce more photosynthates, suggesting compensatory responses modulated by clonal integration (Liu and Dong, 2017).

The asexual reproduction (indicated by ramet number) and main stem length of the apical ramets were higher or marginally higher in the connected than in the disconnected treatment after submergence, suggesting clonal integration promoted morphological plasticity upon submergence. Other studies have also shown shoot plasticity of this species upon submergence, e.g., stem elongation (Fan et al., 2015; Luo et al., 2011; Zhang et al., 2016); however, the elongation is often caused by increasing internode length rather than ramet number (Fan et al., 2015; Luo et al., 2011). In this study, clonal integration enhanced stem elongation of the submerged ramets via increase of ramet number, which needs to consume more energy compared with internode elongation (Voesenek et al., 2006), and further suggests resources translocation from un-submerged ramets (Liu and Dong, 2017; Wang et al., 2017a,b).

Performance of the whole fragments is determined by the integrative performance of both apical and basal ramets (Song et al., 2013). The benefits of clonal integration to both apical and basal ramets significantly increased the growth of the whole fragments. Our results are consistent with the findings of a meta-analytical study involving 60 clonal plant species (Song et al., 2013) and suggest that clonal integration can greatly improve population expansion of *A. philoxeroides*

from un-submerged to submerged conditions. Correspondingly, many studies have shown *A. philoxeroides* could successfully invade into wetland communities, displaying vigorous growth and high morphological plasticity even under unfavorable conditions (Pan et al., 2007; Wang et al., 2009; Zhang et al., 2016). The high invasiveness of this species in wetlands may be partly due to the aid of clonal integration between interconnected ramets (Dong et al., 2015; Liu and Dong, 2017; Luo et al., 2014; Wang et al., 2009).

Performance of de-submerged plants during recovery phase has been considered to be important for evaluating their real tolerance to submergence (Striker, 2012; Voesenek and Bailey-Serres, 2015). De-submergence may impose great stress on plants (Blokhina et al., 2003; Colmer and Voesenek, 2009; Santosa et al., 2007; Voesenek and Bailey-Serres, 2015), e.g., some species which grow vigorously during submergence may have low survival rate or slow recovery of growth after de-submergence (van Eck et al., 2004; Striker, 2012). Therefore, we expected that clonal integration may also play a role during de-submergence. Contrary to our expectation, we found that clonal integration had no significant effect on growth, photosynthetic capacities or asexual reproduction of *A. philoxeroides* after 20 days of de-submergence. One possible reason is that the photosynthetic apparatus of the apical ramets of both connected and disconnected fragments was not severely impaired during submergence (Fv/Fm > 0.7), and Fv/Fm and ΔF/Fm′ of the apical ramets with stolon connection fully recovered to the level of un-submerged basal ramets after de-submergence, increasing by 6% and 11%, respectively. Former studies have also found fast recovery of photosynthetic capacities of this species, i.e., in a few days after submergence (Luo et al., 2011). The quickly recovered photosynthetic capacities of de-submerged apical ramets may well support their recovery growth, therefore leading to weak translocation of resources between the apical and the basal ramets (Liu et al., 2016; Yu et al., 2001). However, for species that cannot quickly recover from de-submergence, clonal integration may play a great role. Therefore, the next critical step is to compare roles of clonal integration between species that can and cannot quickly recover from de-submergence.

We conclude that clonal integration significantly increased performance of submerged ramets of *A. philoxeroides* with the aid of un-submerged ramets, but played little role in its performance during de-submergence, probably due to quickly recovered photosynthetic capacities. However, strong decrease in hydraulic conductivity of roots or severe damage of photosynthetic apparatus after de-submergence is common for many riparian species (Holbrook and Zwieniecki, 2003), which may cause wilting of large proportions of shoot organs or delayed recovery of photosynthetic capacities. For these species, translocation of photosynthates by clonal integration from un-submerged ramets to formerly submerged ramets may be important for their growth recovery. Future studies should be conducted to test this hypothesis.

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